ASSETTO CORSA TRACK CREATION PIPELINE

Complete Workflow from Terrain to Final Validation

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GEOMETRIC CONSTRAINTS & PROJECT SETUP

Reference Standard: Nürburgring Nordschleife

- **Length**: ~20.8 km
- **Real footprint**: ~18×18 km
- **Working bounding box**: [redacted] (coordinates XYZ from [redacted] to [redacted] in Blender/AC units, where [redacted])

Mandatory Rules

- All elements (terrain, splines, objects, run-offs) MUST be contained within this bounding box
- Before export: [redacted] in Blender
- Verify bounding boxes in [redacted] or [redacted]

CHAPTER 1: TERRAIN LAYER

Objective

Create the base heightmap on which all other layers will be placed.

Workflow

1. Generate heightmap using [redacted], [redacted], or Blender (plugin [redacted])
2. **Recommended resolution**: [redacted] for a 20×20 km square (≈5m/pixel, scalable with lateral splines)
3. Import into [redacted] → [redacted] → assign base material (grass/rock)
4. Level critical areas (start/finish line, pit entry/exit, main run-offs)
5. Export as [redacted] or integrate into track package

Assetto Corsa Notes

- Avoid slopes >45° (not supported by physics engine)
- Maintain continuity between terrain chunks to prevent invisible "cracks"
- Use proper UV scaling for texture repetition

CHAPTER 2: VEGETATION

Objective

Define the biome and optimize draw calls.

Workflow

1. Use optimized [redacted] assets or generate with [redacted]/[redacted] + [redacted]
2. Distribute via [redacted] or scattering scripts
3. **Respect clearance zones**:
   - [redacted] from track edge
   - [redacted] from barriers
4. Apply LOD (3-4 levels), alpha-clip for leaves, simplified collisions ([redacted] mesh)
5. Export as [redacted] (or split by quadrants if >500k polygons)

Assetto Corsa Notes

- Limit polygons/visible to [redacted] per chunk
- Use [redacted] for grass/tree transitions
- Avoid PNG textures (performance killer) - use DDS format

CHAPTER 3: TRACK LAYOUT

Objective

Draw the main spline and generate asphalt surface.

Workflow

1. Draw centerline in Blender/CAD → export [redacted] or [redacted]
2. Import into [redacted] or [redacted]
3. **Set parameters**:
   - Width: [redacted] for GT3, [redacted] for formula cars
   - Banking and variable camber
   - Nodes every [redacted]
4. Generate track mesh + kerbs with differentiated materials
5. Validate spline continuity, no intersections, correct normal orientation
6. Export as [redacted]

Assetto Corsa Notes

- Maintain consistent [redacted] for asphalt textures
- Avoid [redacted] in nodes to prevent physics clipping
- Use proper curb heights ([redacted] for standard kerbs)

CHAPTER 4: PIT LANE

Objective

Create functional pit lane compliant with AC rules.

Workflow

1. Pit entry/exit splines connected to main track (angle ≤30° for smooth transition)
2. **Width**: [redacted], speed limit configurable in [redacted]
3. Add: entry/exit lines, timing loops, signage, garages (static or LOD)
4. Configure [redacted] and [redacted] in [redacted] files
5. Test with AI and limiter in-game; verify cars don't slide out of lane

Assetto Corsa Notes

- Use [redacted] in materials
- Position [redacted] and [redacted] with precision ±0.5 m
- Ensure pit lane doesn't interfere with racing line

CHAPTER 5: RUN-OFF AREAS

Objective

Ensure physical and visual safety without penalizing performance.

Workflow

1. Extrude geometries beyond track edge ([redacted] based on corner speed)
2. Assign correct materials: [redacted], [redacted], [redacted] → mapped in [redacted]
3. Insert barriers: [redacted], [redacted], [redacted] with collision mesh [redacted]
4. Model gradual transitions (e.g., asphalt → gravel → grass) to prevent abnormal bouncing
5. Optimize: unify meshes where possible, use [redacted] for high complexity

Assetto Corsa Notes

- Run-offs affect [redacted], [redacted], [redacted]
- Test with [redacted] physics debug mode
- Avoid abrupt height changes (>0.3 m)

CHAPTER 6: BUILDINGS

Objective

Enrich visual context without impacting frustum culling.

Workflow

1. Model: grandstands, control tower, paddock, warehouses, historic structures
2. Real scale, clean UV unwrap, PBR materials ([redacted], [redacted], [redacted], [redacted])
3. **Mandatory LODs** (High/Med/Low), static light baking if needed
4. Strategic placement: landmarks for drivers, never in direct line with high-speed corners
5. Export as [redacted] (or split by zone)

Assetto Corsa Notes

- Max [redacted] total for simultaneously visible buildings
- Use [redacted] manually if supported by CSP
- Keep texture resolution at [redacted] or [redacted] max

CHAPTER 7: EMERGENCY VEHICLES

Objective

Position static/AI vehicles for realism and functionality.

Workflow

1. Select/convert models: `medical car`, `tow truck`, `fire engine` (`.kn5` format)
2. Position in: pit lane, critical run-offs, marshal posts, technical areas
3. Add light collisions, emergency lights (optional via `extra_fx`)
4. If dynamic: configure spawn points in `ai.ini` or `extra_spawns.ini`
5. Verify scale (1:1) and ground alignment (`snap to ground`)

Assetto Corsa Notes

- Avoid interpenetrating meshes with barriers
- Use correct `shadow_bias` to prevent artifacts
- Limit to 5-10 visible vehicles simultaneously

CHAPTER 8: MARSHALS

Objective

Staff critical zones with personnel and signage.

Workflow

1. Posts every `50-80 m` in blind corners, heavy braking zones, narrow run-offs
2. Add: flags, signs, lighting, static/animated figures (CSP supports `marshal_ai`)
3. Configure visibility: avoid clutter, maintain `clear line of sight` to track
4. Export as `marshals.kn5` or integrate into `objects.ini`

Assetto Corsa Notes

- Use `alpha_test` for fences
- Limit textures to `1024×1024` for distance detail
- Position at least `2 m` from track edge

CHAPTER 9: SECONDARY ROADS

Objective

Connect technical areas, parking, external access without interfering with racing.

Workflow

1. Trace service splines ([redacted] width), away from racing line ([redacted])
2. Apply signage, stripes, speed bumps, gates if needed
3. Connect to: pit lane, buildings, marshal zones, vehicle parking
4. Configure as [redacted] or [redacted] in [redacted]
5. Test AI pathfinding for support vehicles

Assetto Corsa Notes

- Avoid unmarked 90° intersections
- Use [redacted] for visual separation
- Never cross the racing line

CHAPTER 10: TRACK CAMERAS

Objective

Configure TV cameras for replays and spectators.

Workflow

1. File Structure

Create [redacted] in [redacted]

Multiple sets: [redacted], [redacted], [redacted], etc.

2. Basic Configuration

ini

```

```

3. Key Parameters

- **POSITION**: XYZ coordinates of camera location
- **FORWARD**: Normalized vector direction camera is pointing
- **UP**: Up vector (usually 0,1,0)
- **MIN_FOV/MAX_FOV**: Field of view range (low values = zoom)
- **IN_POINT/OUT_POINT**: Position on track spline (0.0-1.0) where camera activates
- **IS_FIXED**: 1 = static camera, 0 = camera follows action

4. Moving Cameras (SPLINE)

```
ini
```

CSV Format (one frame per line):

- Create CSV file with frame-by-frame positions
- Export from 3D software (Blender/3ds Max/Maya)
- **SPLINE_ROTATION**: Counter-clockwise rotation in degrees (0° = +X east, 270° = -Z north)
- **SPLINE_ANIMATION_LENGTH**: Lower = faster movement (doesn't follow cars)

5. Strategic Placement

- **Main straight** (start/finish)
- **Critical corners** (Turn 1, hairpins)
- **Overtaking zones** (DRS areas, chicanes)
- **Pit lane entry/exit**
- **Grandstands and scenic viewpoints**

6. Camera Sets

- **Set 1** ([redacted]): Standard trackside cameras
- **Set 2** ([redacted]): Low-angle cameras
- **Set 3** ([redacted]): Aerial/drone cameras
- Switch between sets in replay bar (F3 in-game)

Assetto Corsa Notes

- Cameras must follow spline progression (increasing IN_POINT)
- For start/finish: use IN_POINT=0.0 to ensure correct activation
- Avoid FOV < 2 or > 90 (rendering issues)
- **FOV_GAMMA**: Keep at 0 for standard interpolation
- Test always in replay with slow and fast cars
- Use [redacted] for static cameras, [redacted] for following cameras
- **NEAR_PLANE/FAR_PLANE**: Control rendering distance (avoid too low values)

CHAPTER 11: SECTOR TIMING & CURVE NAMES

Objective

Define timing sectors and name corners for telemetry and apps.

Workflow

1. Sector Timing Gates

Place invisible objects in 3D scene:

- [redacted] - Sector 1/2 boundary (at ~33% of lap)
- [redacted] - Sector 2/3 boundary (at ~66% of lap)
- [redacted] - (Optional) for 4+ sectors

Sector Structure:

1

2. Placement Rules

- Gates must be **perpendicular** to racing direction
- Position at clear, straight sections (not mid-corner)
- Avoid placing in pit entry/exit zones
- Height: [redacted] above track surface
- Width: span entire track + run-offs

3. Curve Naming (Optional but Recommended)

Method A: Simple INI File Create [redacted] in [redacted] folder:

```ini
```

Method B: UI Track JSON Edit `ui_track.json`:

```json
```

4. App Integration

- Apps like "Track Sectors" read `ac_time_X` gates
- Curve names display in telemetry (CSP compatible)
- Configure in `ui_track.json` for interactive maps
- Support for Custom Shaders Patch (CSP) v2.2+

5. Validation

- Test with "Delta Timing" or "Sector Timer" app in-game
- Verify sectors are proportional (not too short/long)
- Check times are recorded correctly in hotlap/qualify
- Ensure gates trigger properly (no double-triggering)

Assetto Corsa Notes

- **For 3 sectors**: Only need `ac_time_0` and `ac_time_1`
- **Point-to-point tracks**: Configure `ac_ab_start` and `ac_ab_end` correctly
- Gates use trigger volumes - cars must pass through to register
- Sector times work in: Practice, Qualifying, Hotlap, Race sessions

- Avoid placing gates too close together (<100 m)

TOOLS & SOFTWARE

Phase	Software/Plugin
3D Modeling	Blender + [redacted] (io_export_kn5), 3ds Max, Maya
Terrain/Heightmap	Gaea, WorldMachine, L3DT, [redacted]
Spline/Track	[redacted], [redacted], [redacted]
Textures/Materials	Substance Painter, GIMP, [redacted], Photoshop
Vegetation	SpeedTree, Forest Pack (3ds Max), [redacted]
Configuration	[redacted], [redacted], Notepad++
Testing	[redacted], [redacted], Polycount Monitor
Camera Animation	Blender (CSV export), 3ds Max (Graph Editor)

FINAL VALIDATION CHECKLIST

Geometric Constraints

- All objects contained within [redacted] bounding box
- Scale applied correctly ([redacted] in Blender)
- No objects outside world bounds

3D Models & Materials

- Normals oriented outward, no visible backfaces
- Collision meshes [redacted] active and non-interpenetrating
- LODs working for vegetation, buildings, vehicles
- Textures in DDS format (no PNG)
- UV mapping correct, no stretching

Configuration Files

- [redacted] complete for all materials (track, pit, run-off, roads)
- [redacted] properly configured
- [redacted] with correct track info and preview
- [redacted] with minimum 5-8 cameras tested
- Sector gates ([redacted], [redacted]) positioned correctly
- Pit lane configuration ([redacted], [redacted]) accurate

Performance

- [redacted] visible simultaneously
- [redacted]
- FPS stable at 60+ on mid-range PC
- No memory leaks after 10+ laps

In-Game Testing

- AI line working (no crashes, proper racing line)
- Pit lane entry/exit functional
- Sector times recording correctly
- Cameras switching properly in replay
- Weather/rain working (if CSP active)
- Night lighting functional (if applicable)
- Multiplayer sync tested
- No clipping or physics glitches

Documentation

- Track name, author, version in [redacted]
- README with installation instructions
- Changelog for updates
- Screenshot for Content Manager preview

OPERATIONAL NOTES

Pipeline Philosophy

- The pipeline is **sequential but iterative**: It's normal to return to terrain or spline after placing run-offs or buildings
- The 20×20 km square is a **project limit**, not a rendering limit: Assetto Corsa handles world streaming well, but respecting the bounding box prevents export errors and AI pathfinding issues

Modern Standards

- **Custom Shaders Patch (CSP)** and **Content Manager** are essential for:
  - Advanced lighting systems
  - Dynamic vegetation
  - Extended physics
  - Camera tools
  - Real-time preview

Common Pitfalls

1. **Forgetting to apply scale** before export → broken collisions
2. **PNG textures** → massive performance hit
3. **Too many polygons in vegetation** → FPS drops
4. **Incorrect IN_POINT values** in cameras → cameras don't trigger
5. **Sector gates in wrong position** → timing errors
6. **Not testing with AI** → undiscovered pathfinding issues

Recommended Workflow Time

- **Terrain**: 2-5 days
- **Track spline & surface**: 3-7 days
- **Vegetation**: 3-10 days
- **Buildings & props**: 5-15 days
- **Cameras & sectors**: 1-3 days
- **Testing & optimization**: 3-7 days
- **Total**: 17-47 days for professional-quality track